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# =====
# Experiment No. 3
# Implementation and Analysis of Spatial Filtering Techniques
# using Low-Pass and High-Pass Filters
# =====

import cv2
import numpy as np
import matplotlib.pyplot as plt

print("Libraries imported successfully!")
Libraries imported successfully!

# =====
# 2. LOAD IMAGE
# =====

image_path = "immm.jpg"

image = cv2.imread(image_path)

if image is None:
    print("ERROR: Image not found!")
else:
    print("Image loaded successfully!")
    print("Image shape:", image.shape)

Image loaded successfully!
Image shape: (495, 660, 3)

# =====
# 3. CONVERT IMAGE TO GRAYSCALE
# =====

gray = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)

print("Grayscale image created successfully!")
print("Grayscale shape:", gray.shape)

Grayscale image created successfully!
Grayscale shape: (495, 660)

# =====
# 4. DISPLAY ORIGINAL IMAGE
# =====

plt.figure(figsize=(6, 5))
plt.imshow(gray, cmap="gray")
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plt.title("Original Grayscale Image")
plt.axis("off")
plt.show()
```

Original Grayscale Image



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# =====
# 5. GAUSSIAN FILTERING
# =====

gaussian = cv2.GaussianBlur(gray, (5, 5), 0)
print("Gaussian filtering completed!")
Gaussian filtering completed!

# =====
# 6. DISPLAY GAUSSIAN FILTERED IMAGE
# =====

plt.figure(figsize=(6, 5))
plt.imshow(gaussian, cmap="gray")
plt.title("Gaussian Filtered Image")
plt.axis("off")
plt.show()
```

Gaussian Filtered Image



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# =====  
# 7. MEDIAN FILTERING  
# =====  
  
median = cv2.medianBlur(gray, 5)  
print("Median filtering completed!")  
Median filtering completed!  
  
# =====  
# 8. DISPLAY MEDIAN FILTERED IMAGE  
# =====  
  
plt.figure(figsize=(6, 5))  
plt.imshow(median, cmap="gray")  
plt.title("Median Filtered Image")  
plt.axis("off")  
plt.show()
```

Median Filtered Image



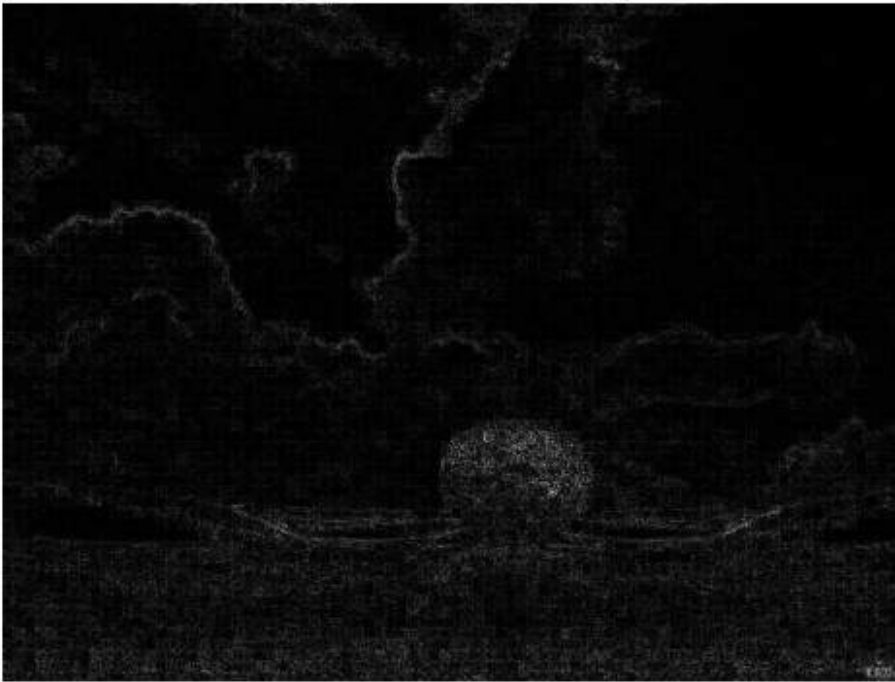
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# =====  
# 9. AVERAGE / MEAN FILTERING  
# =====  
  
average = cv2.blur(gray, (5, 5))  
  
print("Average filtering completed!")  
Average filtering completed!  
  
# =====  
# 10. DISPLAY AVERAGE FILTERED IMAGE  
# =====  
  
plt.figure(figsize=(6, 5))  
plt.imshow(average, cmap="gray")  
plt.title("Average Filtered Image")  
plt.axis("off")  
plt.show()
```

Average Filtered Image



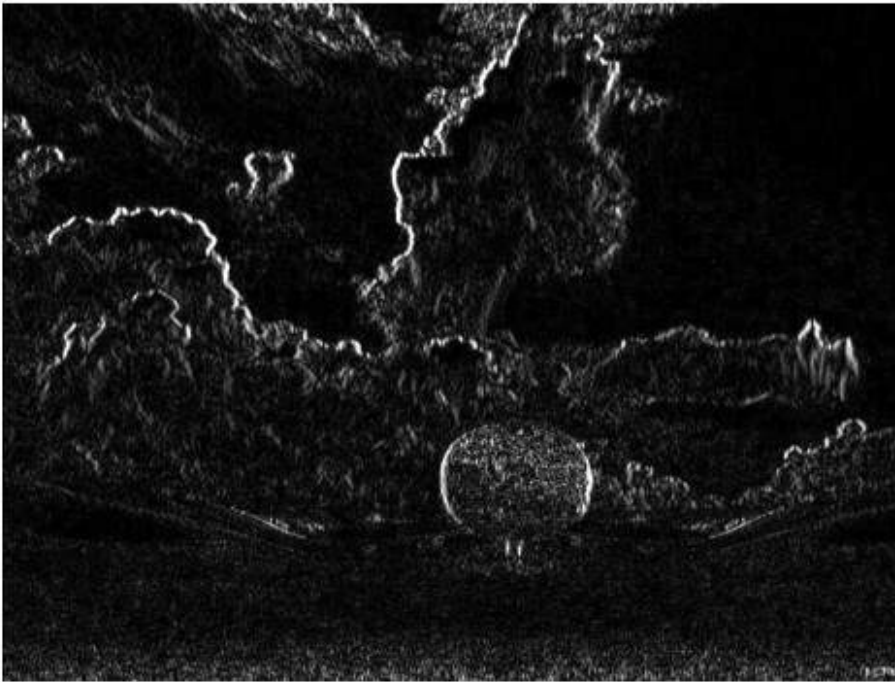
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# =====  
# 11. LAPLACIAN FILTERING  
# =====  
  
laplacian = cv2.Laplacian(gray, cv2.CV_64F)  
laplacian = cv2.convertScaleAbs(laplacian)  
print("Laplacian filtering completed!")  
Laplacian filtering completed!  
  
# =====  
# 12. DISPLAY LAPLACIAN IMAGE  
# =====  
  
plt.figure(figsize=(6, 5))  
plt.imshow(laplacian, cmap="gray")  
plt.title("Laplacian Filtered Image")  
plt.axis("off")  
plt.show()
```

Laplacian Filtered Image



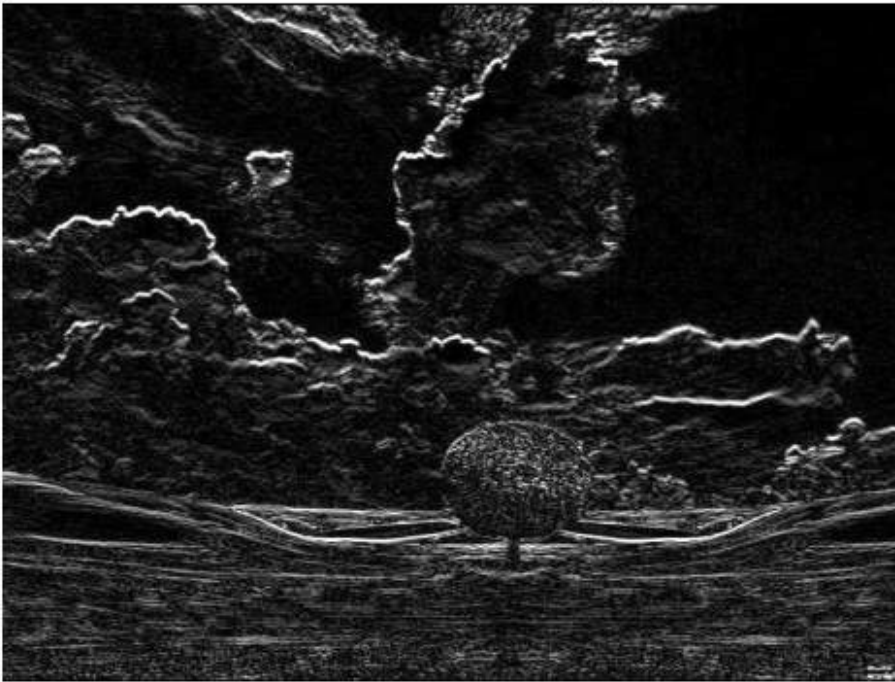
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# =====  
# 13. SOBEL X FILTERING  
# =====  
  
sobel_x = cv2.Sobel(  
    gray,  
    cv2.CV_64F,  
    1,  
    0,  
    ksize=3  
)  
  
sobel_x = cv2.convertScaleAbs(sobel_x)  
print("Sobel X filtering completed!")  
Sobel X filtering completed!  
  
# =====  
# 14. DISPLAY SOBEL X IMAGE  
# =====  
  
plt.figure(figsize=(6, 5))  
plt.imshow(sobel_x, cmap="gray")  
plt.title("Sobel X - Horizontal Edges")  
plt.axis("off")  
plt.show()
```

Sobel X - Horizontal Edges



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# =====  
# 15. SOBEL Y FILTERING  
# =====  
  
sobel_y = cv2.Sobel(  
    gray,  
    cv2.CV_64F,  
    0,  
    1,  
    ksize=3  
)  
  
sobel_y = cv2.convertScaleAbs(sobel_y)  
print("Sobel Y filtering completed!")  
Sobel Y filtering completed!  
  
# =====  
# 16. DISPLAY SOBEL Y IMAGE  
# =====  
  
plt.figure(figsize=(6, 5))  
plt.imshow(sobel_y, cmap="gray")  
plt.title("Sobel Y - Vertical Edges")  
plt.axis("off")  
plt.show()
```

Sobel Y - Vertical Edges



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# =====  
# 17. COMPARE LOW-PASS FILTERS  
# =====  
  
plt.figure(figsize=(15, 10))  
  
plt.subplot(2, 2, 1)  
plt.imshow(gray, cmap="gray")  
plt.title("Original")  
plt.axis("off")  
  
plt.subplot(2, 2, 2)  
plt.imshow(gaussian, cmap="gray")  
plt.title("Gaussian Filter")  
plt.axis("off")  
  
plt.subplot(2, 2, 3)  
plt.imshow(median, cmap="gray")  
plt.title("Median Filter")  
plt.axis("off")  
  
plt.subplot(2, 2, 4)  
plt.imshow(average, cmap="gray")  
plt.title("Average Filter")  
plt.axis("off")
```

```
plt.tight_layout()
plt.show()
```



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# =====
# 18. COMPARE HIGH-PASS FILTERS
# =====

plt.figure(figsize=(15, 10))

plt.subplot(2, 2, 1)
plt.imshow(gray, cmap="gray")
plt.title("Original")
plt.axis("off")

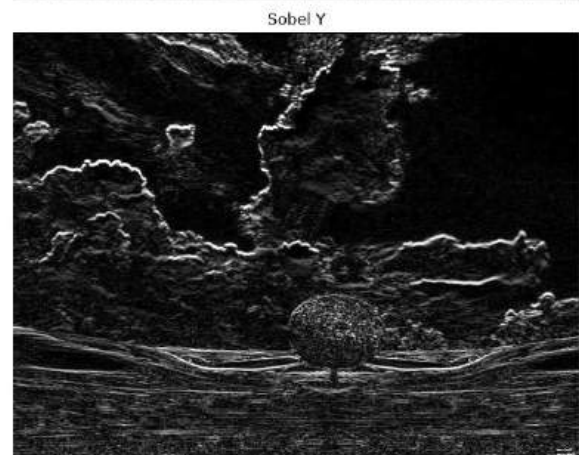
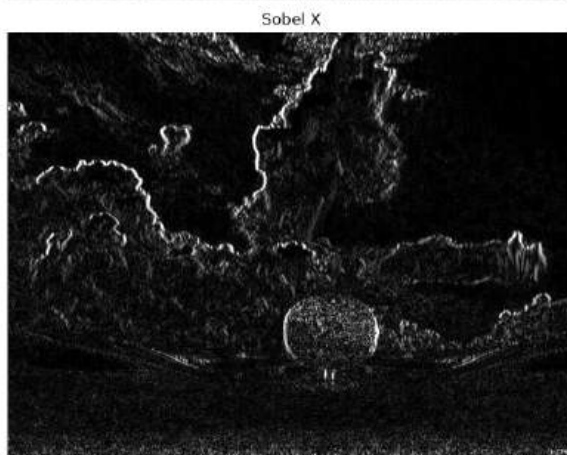
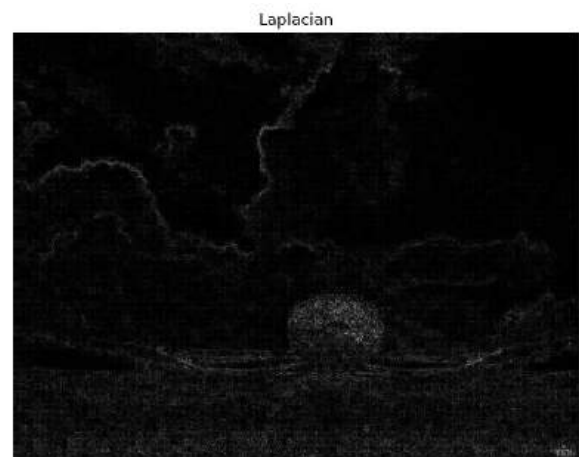
plt.subplot(2, 2, 2)
plt.imshow(laplacian, cmap="gray")
plt.title("Laplacian")
plt.axis("off")

plt.subplot(2, 2, 3)
plt.imshow(sobel_x, cmap="gray")
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plt.title("Sobel X")
plt.axis("off")

plt.subplot(2, 2, 4)
plt.imshow(sobel_y, cmap="gray")
plt.title("Sobel Y")
plt.axis("off")

plt.tight_layout()
plt.show()
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# =====
# 19. FINAL COMPARISON OF ALL FILTERS
# =====

plt.figure(figsize=(18, 10))

plt.subplot(2, 4, 1)
plt.imshow(gray, cmap="gray")
plt.title("Original")
plt.axis("off")
```

```
plt.subplot(2, 4, 2)
plt.imshow(gaussian, cmap="gray")
plt.title("Gaussian")
plt.axis("off")

plt.subplot(2, 4, 3)
plt.imshow(median, cmap="gray")
plt.title("Median")
plt.axis("off")

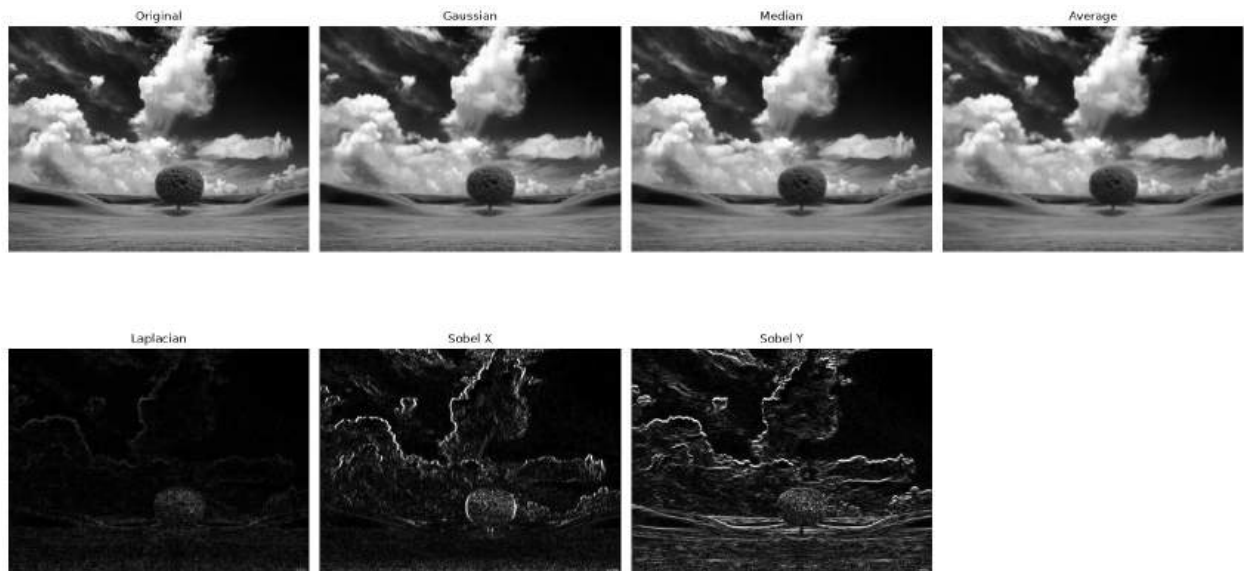
plt.subplot(2, 4, 4)
plt.imshow(average, cmap="gray")
plt.title("Average")
plt.axis("off")

plt.subplot(2, 4, 5)
plt.imshow(laplacian, cmap="gray")
plt.title("Laplacian")
plt.axis("off")

plt.subplot(2, 4, 6)
plt.imshow(sobel_x, cmap="gray")
plt.title("Sobel X")
plt.axis("off")

plt.subplot(2, 4, 7)
plt.imshow(sobel_y, cmap="gray")
plt.title("Sobel Y")
plt.axis("off")

plt.tight_layout()
plt.show()
```



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# =====
# 20. SAVE FILTERED IMAGES
# =====

cv2.imwrite("gaussian.jpg", gaussian)
cv2.imwrite("median.jpg", median)
cv2.imwrite("average.jpg", average)
cv2.imwrite("laplacian.jpg", laplacian)
cv2.imwrite("sobel_x.jpg", sobel_x)
cv2.imwrite("sobel_y.jpg", sobel_y)

print("Images saved successfully!")
print("1. gaussian.jpg")
print("2. median.jpg")
print("3. average.jpg")
print("4. laplacian.jpg")
print("5. sobel_x.jpg")
print("6. sobel_y.jpg")

Images saved successfully!
1. gaussian.jpg
2. median.jpg
3. average.jpg
4. laplacian.jpg
5. sobel_x.jpg
6. sobel_y.jpg

# =====
# 21. EXPERIMENT COMPLETED
# =====

print("=====")
```

```
print("Experiment No. 3 completed successfully!")
print("Spatial filtering techniques implemented:")
print("1. Gaussian Filter")
print("2. Median Filter")
print("3. Average Filter")
print("4. Laplacian Filter")
print("5. Sobel X Filter")
print("6. Sobel Y Filter")
print("=====")
```

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=====
Experiment No. 3 completed successfully!
Spatial filtering techniques implemented:
1. Gaussian Filter
2. Median Filter
3. Average Filter
4. Laplacian Filter
5. Sobel X Filter
6. Sobel Y Filter
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```